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RESEARCH ARTICLE

Fractional Kaiser Window With Application to Finite Impulse Response Digital Filter Design

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ABSTRACT This study introduces a new adjustable window derived from the recently proposed fractional modified Bessel function of the first kind with zero order. The proposed window, referred to as the fractional Kaiser window in this study, is obtained using the Kaiser approach, which was originally used to develop the well-known Kaiser window, and encompasses the Kaiser window as a special case. First, spectral analysis of the proposed window is conducted across various window lengths to evaluate its spectral performance. The comparative results reveal that the proposed window offers superior ripple ratio values, ranging from 1.67 dB to 5.20 dB, compared to the Kaiser window for the same half-mainlobe width at $N = 51$. Next, the effectiveness of the proposed window in the design of finite impulse response (FIR) digital filters is assessed. After analyzing the filter's spectral performance for different values of the independent parameters, a general comparison is made with filters designed using the Kaiser window, demonstrating that superior filters can be achieved with the proposed window. Finally, three design examples of digital filters are provided, with different filter lengths ($N = 51$, 71, and 101) and filter transition widths, to compare with other windows in the literature. The simulation results indicate that the proposed window consistently produces better filter designs in all three examples by offering the largest minimum stopband attenuation values.

INDEX TERMS Digital filter, FIR filter, filter design, fractional Kaiser window, window function, windowing.

I. INTRODUCTION

Digital filters are systems or algorithms designed to process a wide range of digital signals, including audio, image, video, communication, and time-series data, to achieve specific effects, such as removing unwanted components or extracting valuable information from the signal. They are classified as finite impulse response (FIR) and infinite impulse response (IIR) according to their impulse response duration. FIR digital filters are often preferred in digital filtering applications where exact phase linearity and stability are required. An FIR filter with a length of N (or order of N - 1) can be expressed by the transfer function as given in (1) [1].

$$H(z) = a_0 + a_1 z^{-1} + a_2 z^{-2} + \ldots + a_{N-1} z^{-(N-1)} \quad (1)$$

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Designing an FIR filter with a length of N requires finding N coefficients (a_0, a_1, a_2,..., a_{N-1}) that will satisfy the prescribed filter spectral characteristics defined generally by the parameters of the minimum stopband attenuation and filter transition width. Optimization methods yield the most superior designs among FIR filter design techniques, and ongoing research continues to enhance these approaches [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13]. However, these optimization-driven methods are computationally intensive, making them unsuitable for applications that require real-time processing. In contrast, the Fourier series method based on window functions truncating and smoothing the ideal filter has a simpler approach for designing FIR filters, requiring significantly less computational effort than other methods [14].

A discrete-time window function is a time function that remains non-zero within a specific time interval (window length N) and zero outside of that interval, as given in (2).

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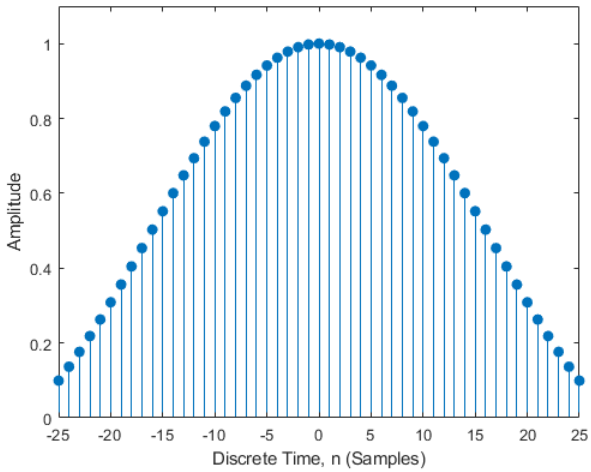

FIGURE 1. A window function plotted for a window length of $N = 51$.

The normalized amplitude characteristic of a discrete-time window is typically depicted in Figure 1.

$$w(n) = \begin{cases} \neq 0 & \text{if } n \leq \dfrac{N-1}{2} \\ = 0 & \text{if } n > \dfrac{N-1}{2} \end{cases} \tag{2}$$

Window functions, or simply windows, are categorized as fixed or adjustable. Fixed windows such as Rectangular, Von Hann, Hamming, and Blackman rely on a single parameter (N) as an independent variable, whereas adjustable windows such as Kaiser, Saramaki, and Dolph-Chebyshev use two or more parameters [1]. In practical applications, adjustable windows are often preferred owing to their high flexibility.

Windowing-based digital filter designs often exhibit undesirable Gibbs oscillations. To minimize these oscillations, many adjustable windows have been introduced in the literature to design FIR digital filters. However, because sub-optimal solutions are provided and best window function depends on the applications, there is always a need to develop new window functions that can deliver improved filter performance. This study aims to propose a new family of adjustable window functions to improve window-based design of digital filters.

The rest of the paper is organized as follows: In Section II, adjustable windows from the literature to be used for the comparison purpose with the proposed window are presented after a brief mention of fixed windows. Section III introduces the proposed window function. In Section IV, the methods for window spectral analysis and filter design using windowing are explained. In Section V, simulation results are provided to demonstrate the performance of the proposed method in window spectral analysis and designing FIR digital filters. Section VI summarizes the results obtained.

II. WINDOW FUNCTIONS IN LITERATURE

A. FIXED WINDOWS

Fixed windows are defined by only the window length (N). Rectangular, Von Hann, Hamming, and Blackman windows are the most known fixed windows. Other notable fixed windows in alphabetic order include Bartlett [15], Bartlett-Hann [16], Blackman-Harris [17], Cos(x) [18], Flat-Top [19], Lanczos [20], Nuttal [21], Parabolic [18], Parzen [22], Riemann [18], Triangular [23], and Welch [24]. Due to having only one independent parameter, using a fixed window in a filter design application results in only one spectral characteristic for a given filter length (N). This limited adaptability makes them less practical for real-world applications compared to adjustable windows.

B. ADJUSTABLE WINDOWS

Since adjustable windows are defined by at least two independent parameters including the window length N, they can provide many different window spectral characteristics with a fixed window length. Therefore, these windows can provide greater flexibility in designing filters compared to fixed windows. In literature, many adjustable windows are available such as Dolph-Chebyshev [25], Kaiser [26], Saramaki [27], Cosh [28], Cosh-Hamming [29], Exponential [30], Exponential-Hamming [31], Gaussian [19], Kaiser-Hamming [32], Kaiser-Von Hann [33], Poisson [24], Hann-Poisson [20], Cauchy [17], Parzen-Cos^6 (πt) [22], Ultraspherical [34], modified Cosh [35], and modified Kaiser [36].

Recent studies continue to focus on improving the effectiveness of window functions across various application areas [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49].

Some well known adjustable window functions, which are used for comparison purpose in this study, are defined and explained below in alphabetical order.

1) COSH WINDOW

This window [28] was derived by using Kaiser approach employed in creating the Kaiser window. While Kaiser window uses the modified Bessel function of the first kind with zero order, Cosh window uses the cosine hyperbolic function. This window is defined as in (3) for $|n| \leq \frac{N-1}{2}$.

$$w(n) = \frac{\cosh(\alpha\sqrt{1-\left(\frac{2n}{N-1}\right)^2})}{\cosh(\alpha)} \tag{3}$$

This window has two adjustable parameters as N and α.

2) COSH-HAMMING WINDOW

This window [29] was proposed to improve the ripple ratio for window spectral and minimum stopband attenuation for filter spectral characteristics of Cosh window by hybridizing the Cosh and Hamming windows. This hybrid window is defined as in (4) for $|n| \leq \frac{N-1}{2}$.

$$w(n) = 0.5\left(\frac{\cosh(\alpha\sqrt{1-\left(\frac{2n}{N-1}\right)^2})}{\cosh(\alpha)} + 0.54 - 0.46cos\frac{2\pi n}{N-1}\right) \tag{4}$$

This window has also two adjustable parameters as N and α.

3) DOLPH-CHEBYSHEV WINDOW

Dolph introduced this window in 1946 [25]. It is best known to offer the narrowest main lobe width for a given ripple ratio level. This window is defined as in (5).

$$w(n) = \frac{1}{N}\left[\frac{1}{r} + 2\sum_{i=1}^{\frac{N-1}{2}} T_{N-1}\left(x_0 \cos\left(\frac{i\pi}{N}\right)\right)\cos\left(\frac{2ni\pi}{N}\right)\right] \tag{5}$$

for $n = 0, 1, 2, \ldots, (N-1)/2$ where

$$r = 10^{-\frac{R}{20}} \tag{6}$$

and

$$x_0 = \cosh\left(\frac{1}{N-1}\cosh^{-1}\left(\frac{1}{r}\right)\right) \tag{7}$$

Function $T_k(x)$ is the kth-order Chebyshev polynomial of the first kind, described as (8).

$$T_k(x) = \begin{cases} \cos(k\cos^{-1}(x)) & \text{if } |x| \leq 1 \\ \cosh(\cosh^{-1}(x)) & \text{if } |x| \geq 1 \end{cases} \tag{8}$$

This window has two adjustable parameters as N and R.

4) EXPONENTIAL WINDOW

This window [30] was derived similar to Cosh window. Instead of the cosine hyperbolic function, exponential function is used. This window is defined as in (9).

$$w(n) = \frac{\exp(\alpha\sqrt{1-\left(\frac{2n}{N-1}\right)^2})}{\exp(\alpha)} \tag{9}$$

This window has two adjustable parameters as N and α.

5) EXPONENTIAL-HAMMING WINDOW

This window [31] was proposed to improve the spectral characteristics of the Exponential window by hybridizing the Exponential and Hamming windows. This hybrid window is defined as in (10).

$$w(n) = 0.5\left(\frac{\exp(\alpha\sqrt{1-\left(\frac{2n}{N-1}\right)^2})}{\exp(\alpha)} + 0.54 - 0.46cos\frac{2\pi n}{N-1}\right) \tag{10}$$

This window has also two adjustable parameters as N and α.

6) GAUSSIAN WINDOW

This window [19] is defined as in (11).

$$w(n) = e^{-\frac{1}{2}\left[\alpha\frac{2n}{N-1}\right]^2} \tag{11}$$

This window has two adjustable parameters as N and α.

7) HANN-POISSON WINDOW

This hybrid window [20] is defined as in (12).

$$w(n) = 0.5\left[1 + \cos\left(\frac{2\pi n}{N-1}\right)\right]e^{\alpha\frac{2|n|}{N-1}} \tag{12}$$

This window has two adjustable parameters as N and α.

8) KAISER WINDOW

In 1966, Kaiser introduced the Kaiser window, which has become one of the most well-known adjustable windows. Kaiser later published the filter design equations for this window in 1974 [26]. The Kaiser window is often preferred over the Dolph-Chebyshev window for FIR filter design due to its superior performance.

This window is defined as in (13).

$$w(n) = \frac{I_0(\alpha\sqrt{1-\left(\frac{2n}{N-1}\right)^2})}{I_0(\alpha)} \tag{13}$$

where $I_0(x)$ is the modified Bessel function of the first kind with zero order having the following simple power series expansion in (14)

$$I_0(x) = 1 + \sum_{k=1}^{\infty}\left[\frac{1}{k!}\left(\frac{x}{2}\right)^k\right]^2 \tag{14}$$

This window has two adjustable parameters as N and α.

9) KAISER-HAMMING WINDOW

This window [32] was proposed to improve the spectral characteristics of the Kaiser window by hybridizing the Kaiser and Hamming windows. This hybrid window is defined as in (15).

$$w(n) = 0.5\frac{I_0(\alpha\sqrt{1-\left(\frac{2n}{N-1}\right)^2})}{I_0(\alpha))} + 0.5(0.54 - 0.46cos\frac{2\pi n}{N-1}) \tag{15}$$

This window has two adjustable parameters as N and α.

10) KAISER-VON HANN WINDOW

This window [33] was also proposed to improve the spectral characteristics of the Kaiser window by hybridizing the Kaiser and Von Hann windows. This hybrid window is defined as in (16).

$$w(n) = 0.5\frac{I_0(\alpha\sqrt{1-\left(\frac{2n}{N-1}\right)^2})}{I_0(\alpha))} + 0.5(0.5 - 0.5cos\frac{2\pi n}{N-1}) \tag{16}$$

This window has two adjustable parameters as N and α.

11) SARAMAKI WINDOW

Saramaki proposed this two-parameter window relying on iterative expressions in the time domain, which was shown to offer better minimum stopband attenuation than the Kaiser window [27]. This window is defined as in (17).

$$w(n) = \frac{\hat{w}(n)}{\hat{w}(0)} \quad |n| \leq \frac{N-1}{2} \tag{17}$$

where

$$\hat{w}(n) = v_0(n) + 2\sum_{k=1}^{\frac{N-1}{2}} v_k(n) \tag{18}$$

$v_k(n)$'s in (18) can be calculated using the following recursion formulas (19), (20), and (21):

$$v_0(n) = \begin{cases} 1 & \text{if } n = 0 \\ 0 & \text{otherwise} \end{cases} \tag{19}$$

$$v_1(n) = \begin{cases} \gamma_s - 1 & \text{if } n = 0 \\ \frac{\gamma_s}{2} & \text{if } |n| = 1 \\ 0 & \text{otherwise} \end{cases} \tag{20}$$

and

$$v_k(n) = \begin{cases} 2(\gamma_s - 1)v_{k-1}(n) - v_{k-2}(n) \\ +\gamma_s\left[v_{k-1}(n-1) + v_{k-1}(n+1)\right] & \text{if } |n| \leq k \\ 0 & \text{otherwise} \end{cases} \tag{21}$$

where γ_s can be defined in terms of β as(22)

$$\gamma_s = \frac{1 + cos\frac{2\pi}{N}}{1 + cos\frac{2\beta\pi}{N}}. \tag{22}$$

This window has two adjustable parameters as N and β.

III. PROPOSED WINDOW: FRACTIONAL KAISER WINDOW

Since the proposed window is derived from a recently published function which is the fractional modified Bessel function of the first kind with zero order, firstly this function will be introduced.

A. FRACTIONAL MODIFIED BESSEL FUNCTION OF THE FIRST KIND

The modified Bessel's differential equation is defined as (23)

$$z^2\frac{d^2y}{dz^2} + z\frac{dy}{dz} - (z^2 + \nu^2)y = 0. \tag{23}$$

where $\nu \in \mathbb{Z}$ is the order of the Bessel function, and z is the argument of the Bessel function.

The modified Bessel's differential equation plays a crucial role in mathematical physics and engineering due to its applicability in problems involving cylindrical or spherical symmetry where the variable of interest (such as temperature, pressure, or potential) exhibits exponential behavior, either decaying or growing, rather than oscillating as in the case of the ordinary Bessel functions.

A solution to the modified Bessel's differential equation is the modified Bessel function of the first kind of order ν defined by (24) [50]

$$I_\nu(z) = \sum_{k=0}^{\infty} \frac{1}{\Gamma(k+1)\Gamma(k+\nu+1)} \left(\frac{z}{2}\right)^{2k+\nu}. \tag{24}$$

where $\Gamma(.)$ is the Euler gamma function which is defined for a complex number z with a positive real part as (25)

$$\Gamma(z) = \int_0^\infty t^{z-1}e^{-t}\,dt. \tag{25}$$

Martin and Estrada [51] extended the traditional modified Bessel function of the first kind, $I_\nu(z)$, to incorporate a fractional parameter γ. They introduced the fractional modified Bessel function of the first kind with fractional parameter γ, $I_\nu^\gamma(z)$, as in (26).

$$I_\nu^\gamma(z) = \sum_{k=0}^{\infty} \frac{1}{\Gamma(\gamma k+1)\Gamma(k+\nu+1)} \left(\frac{z}{2}\right)^{2k+\nu}. \tag{26}$$

B. FRACTIONAL KAISER WINDOW

Kaiser window is derived from the modified Bessel function of the first kind with zero order given by (14). This function, $I_0(.)$, can be described in terms of $\Gamma(.)$ function as in (27).

$$I_0(z) = \sum_{k=0}^{\infty} \frac{1}{\Gamma(k+1)\Gamma(k+1)} \left(\frac{z}{2}\right)^{2k}. \tag{27}$$

By replacing $\nu = 0$ for (26), the fractional modified Bessel function of the first kind with zero order becomes (28).

$$I_0^\gamma(z) = \sum_{k=0}^{\infty} \frac{1}{\Gamma(\gamma k+1)\Gamma(k+1)} \left(\frac{z}{2}\right)^{2k}. \tag{28}$$

It is easily seen that the corresponding function of the fractional modified Bessel function of the first kind with zero order for $\gamma = 1$ is indeed the modified Bessel function of the first kind with zero order. In Figure 2, the fractional modified Bessel function of the first kind with zero order is plotted for different γ values.

As the core part of this study, by using the Kaiser approach employed to obtain the Kaiser window, a new window function based on the fractional modified Bessel function of the first kind with zero order is proposed as (29).

$$w(n) = \frac{I_0^\gamma\left(\alpha\sqrt{1 - \left(\frac{2n}{N-1}\right)^2}\right)}{I_0^\gamma(\alpha)}. \tag{29}$$

So, the proposed window, namely fractional Kaiser window, has three independent parameters which are N (window length), α (shape parameter), and γ (fractional parameter). For $\gamma = 1$, the proposed window becomes the traditional Kaiser window. Note that the proposed window for $\gamma = 1$ was numerically compared with the Kaiser window in MATLAB,

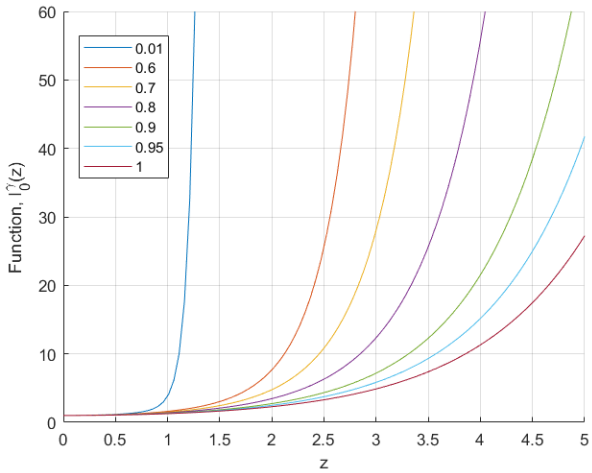

FIGURE 2. Plots of the fractional modified bessel function of the first kind with zero order for different γ values.

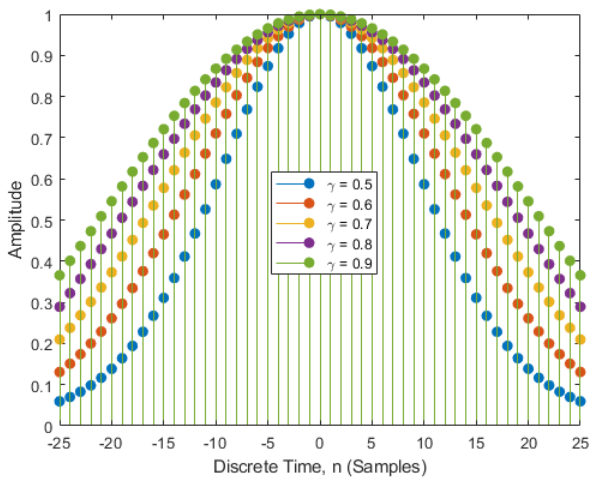

FIGURE 3. Plots of the fractional kaiser window for different γ values at $N = 51$ and $\alpha = 2$.

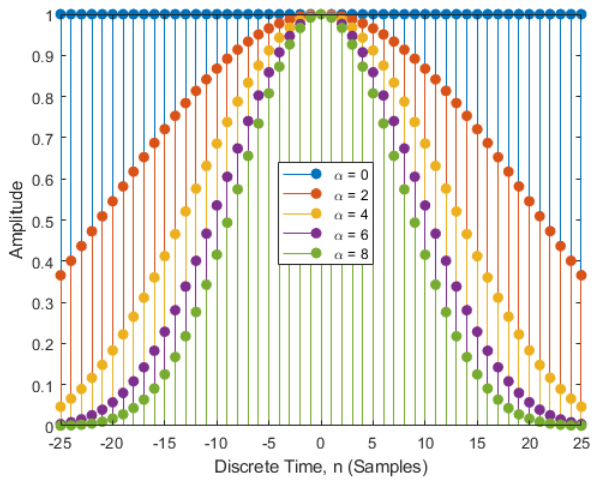

FIGURE 4. Plots of the fractional kaiser window for different α values at $N = 51$ and $\gamma = 0.9$.

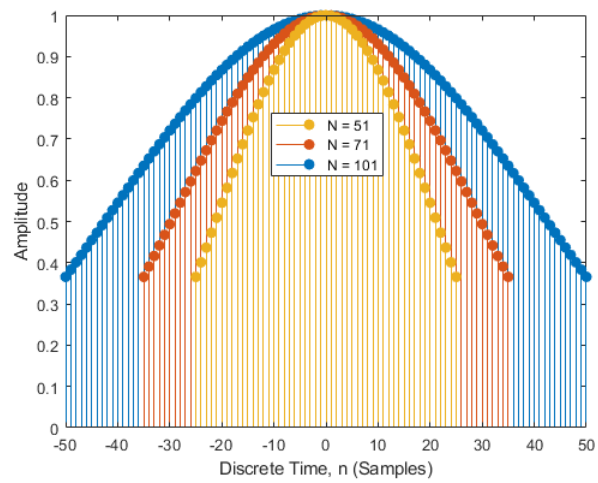

FIGURE 5. Plots of the fractional kaiser window for different N values at $\alpha = 2$ and $\gamma = 0.9$.

and it was found that using 20 terms in the summation given by (28) is sufficient, rather than extending the series to infinity. Consequently, all computations in this study are performed using 20 terms for the proposed window.

Figure 3, Figure 4, and Figure 5 illustrate the effects of the independent parameters of the proposed window in time domain. In Figure 3, the fractional Kaiser window is plotted in time domain for different γ values at fixed values of N and α. It is seen that a wider window is obtained as γ increases. As for Figure 4, it is plotted for different α values at fixed values of N and γ, and the figure shows that a narrower window is obtained as α increases. And, Figure 5 shows its characteristics plotted for different window lengths N at fixed values of α and γ. It can be observed that a wider window is obtained as N increases due to increased number of samples.

IV. METHODS

A. WINDOW SPECTRAL CHARACTERISTICS

Performance of the window functions are generally compared in terms of their spectral characteristics. The spectral representation of a window function is given by (30).

$$W(e^{j\omega}) = \sum_{n=-(N-1)/2}^{(N-1)/2} w(n)e^{-j\omega n} \tag{30}$$

Rearranging (30) for the symmetric windows, the frequency spectrum in terms of amplitude ($A(\omega)$) and phase spectrum ($\theta(\omega)$) becomes (31).

$$W\left(e^{j\omega}\right) = A(\omega)\, e^{j\theta(\omega)} = w(0) + 2\sum_{n=1}^{(N-1)/2} w(n)cos(\omega n) \tag{31}$$

By using (32), normalized amplitude spectrum of a window function can be obtained as shown in Figure 6.

$$|W_N(e^{j\omega T})| = 20\log_{10}(\frac{|A(\omega)|}{|A(0)|}) \tag{32}$$

In Figure 6, S_{max} (Maximum sidelobe amplitude in dB), S_{min} (Minimum sidelobe amplitude in dB), ripple ratio

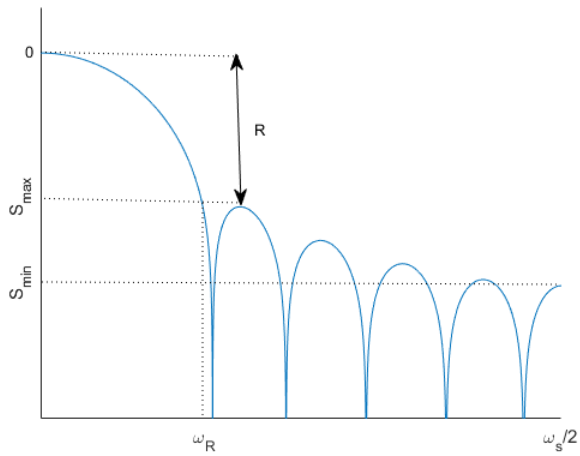

FIGURE 6. Normalized logarithmic amplitude spectrum of a typical symmetric window.

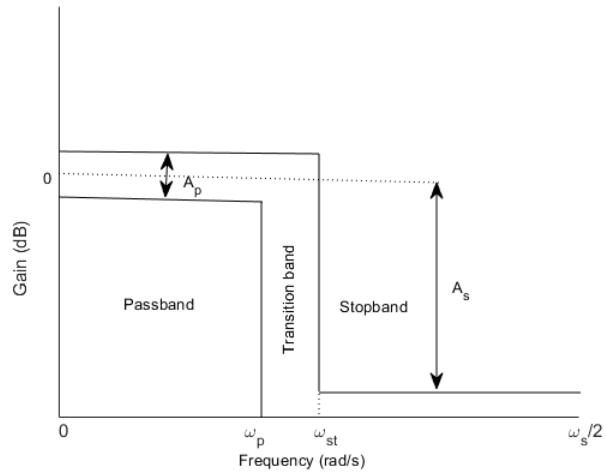

FIGURE 7. Normalized amplitude spectrum of a low-pass filter with its filter spectral parameters.

($R = S_{max}$ - Mainlobe amplitude in dB), half mainlobe width (ω_R), and sampling frequency (ω_s) are shown. Many window spectral parameters are defined for different applications [18].

To distinguish the performance of the window functions for filter design, half mainlobe width (ω_R) and ripple ratio (R) are mostly used among the window spectral characteristic parameters. For FIR digital filter design applications, the mainlobe width is related with the transition width between the passband and stopband whereas the ripple ratio is about the ripples in the passband and stopband. A good window function has a narrower mainlobe width and smaller ripple ratio.

B. WINDOWING BASED DESIGN OF FIR DIGITAL FILTERS

The Fourier series using windowing method is an approach used in designing FIR digital filters. In this method, a window function is employed to truncate and smooth the ideal filter, which has an infinite impulse response which cannot be physically realized. To obtain finite impulse response of a non-causal filter, infinite impulse response is windowed as in (33) [1].

$$h_{nc}(n) = w(n)h_{id}(n). \tag{33}$$

where $h_{id}(nT)$ represents the ideal filter infinite impulse response. An ideal low-pass filter has an infinite impulse response given by (34) [1].

$$h_{\mathrm{id}}(n) = \begin{cases} \dfrac{\omega_c}{\pi} & \text{if } n = 0 \\ \dfrac{\sin(\omega_c n)}{n\pi} & \text{if } n \neq 0 \end{cases} \tag{34}$$

Figure 7 illustrates the normalized amplitude spectrum of a low-pass filter, showing its spectral parameters: passband frequency (ω_p), stopband frequency (ω_{st}), sampling frequency (ω_s), maximum pass-band attenuation (A_p), and minimum stopband attenuation (A_s).

In addition to these parameters, cut-off frequency (ω_c) and transition width ($\Delta\omega$) parameters, defined by (35) and (36) respectively, are also used as filter spectral parameters.

$$\omega_c = \frac{\omega_{st} + \omega_p}{2}. \tag{35}$$

$$\Delta\omega = \omega_{st} - \omega_p. \tag{36}$$

The Gibbs oscillations in both the passband and stopband regions for the digital filters designed by using window functions are approximately equivalent. Thus, when evaluating the performance of the digital filters, calculating only one attenuation parameter (typically A_s) related with the oscillations in these regions is sufficient.

In this study, in evaluating the performance of the designed digital filters, the minimum stopband attenuation values are compared by fixing the filter length and transition width for all filters to have a fair comparison. A filter having the largest minimum stopband attenuation is said to be the best one in this case.

V. SIMULATION RESULTS

A. WINDOW SPECTRAL ANALYSIS FOR THE PROPOSED WINDOW WITH COMPARISON TO THE KAISER WINDOW

To examine how the independent parameters affect the spectral characteristics of the proposed window, spectral analyses are conducted using different values for the parameters N, α, and γ.

First, for a fixed value of γ, the window spectrums are obtained for different α values at various window lengths of $N = 51$, 71, and 101 as illustrated in Figure 8, Figure 9, and Figure 10, respectively. The window spectral parameters from these figures are summarized in Table 1. It can be observed that as α increases, the main lobe width expands while the ripple ratio decreases for each window length.

Secondly, with α held constant, the window spectrums are generated for different γ values at window lengths of $N = 51$, 71, and 101 as illustrated in Figure 11, Figure 12, and Figure 13, respectively. The window spectral parameters from these figures are summarized in Table 2. It is seen that

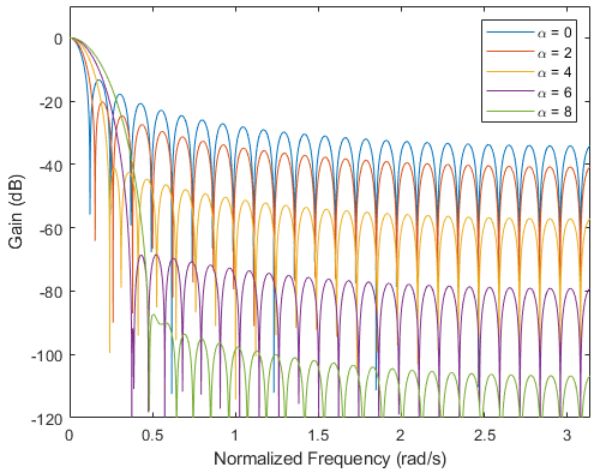

FIGURE 8. Window spectrums of the proposed window for different α values for $N = 51$ and $\gamma = 0.9$.

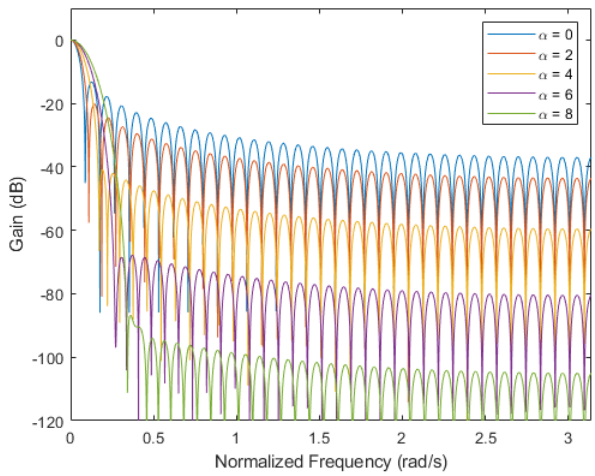

FIGURE 9. Window spectrums of the proposed window for different α values for $N = 71$ and $\gamma = 0.9$.

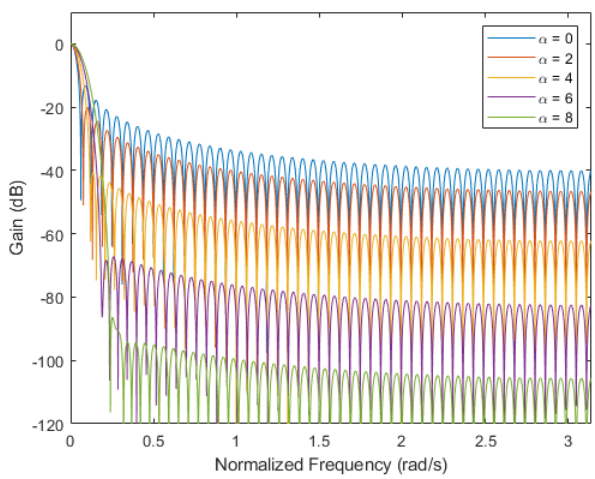

FIGURE 10. Window spectrums of the proposed window for different α values for $N = 101$ and $\gamma = 0.9$.

increasing γ leads to a narrower mainlobe width and a higher ripple ratio for each window length.

TABLE 1. Window spectral values for the proposed window at different window lengths and α values for $\gamma = 0.9$.

	N = 51		N = 71		N = 101	
α	ω_R (rad/s)	R (dB)	ω_R (rad/s)	R (dB)	ω_R (rad/s)	R (dB)
0	0.1001	-13.25	0.0719	-13.26	0.0506	-13.26
2	0.1362	-20.27	0.0975	-20.17	0.0683	-20.10
4	0.2323	-41.24	0.1654	-40.87	0.1155	-40.59
6	0.3611	-68.55	0.2580	-67.83	0.1807	-67.23
8	0.4744	-87.30	0.3384	-86.78	0.2366	-86.33

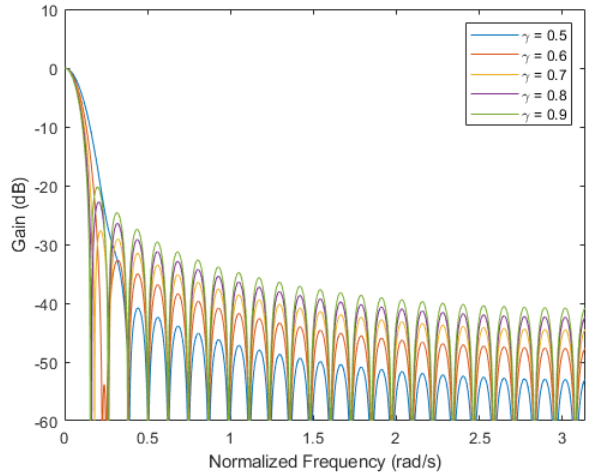

FIGURE 11. Window spectrums of the proposed window for different γ values for $N = 51$ and $\alpha = 2$.

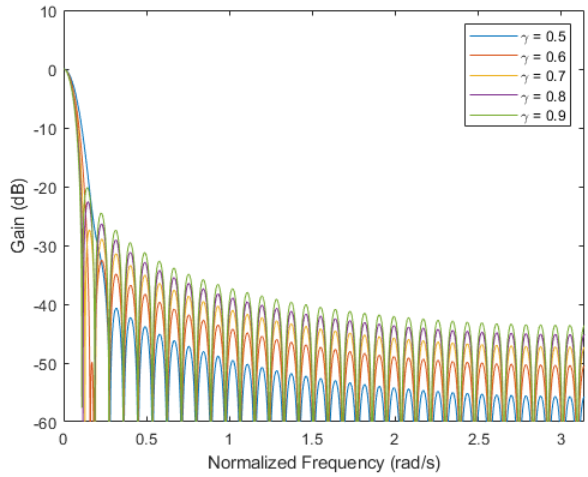

FIGURE 12. Window spectrums of the proposed window for different γ values for $N = 71$ and $\alpha = 2$.

TABLE 2. Window spectral values for the proposed window at different window lengths and γ values for $\alpha = 2$.

	N = 51		N = 71		N = 101	
γ	ω_R (rad/s)	R (dB)	ω_R (rad/s)	R (dB)	ω_R (rad/s)	R (dB)
0.5	0.3617	-40.82	0.2593	-40.67	0.1821	-40.56
0.6	0.2031	-32.68	0.1447	-32.55	0.1011	-32.44
0.7	0.1673	-27.65	0.1195	-27.39	0.0836	-27.20
0.8	0.1477	-22.77	0.1056	-22.63	0.0740	-22.52
0.9	0.1362	-20.27	0.0975	-20.17	0.0683	-20.09

To have a better idea for the usefulness of the new independent parameter, γ, on the window spectral characteristics, the relations between ripple ratio and half mainlobe width for the proposed window for different γ values are obtained

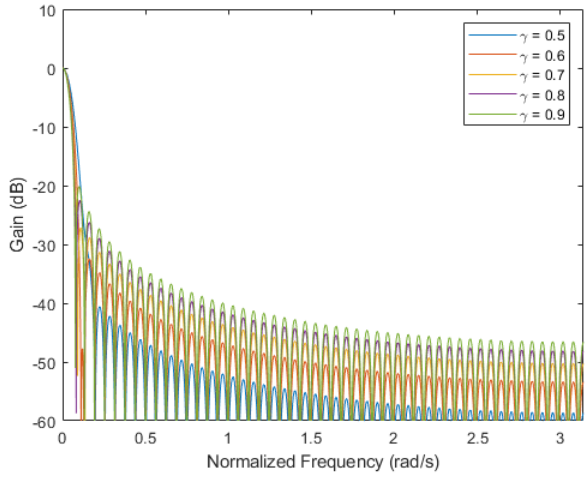

FIGURE 13. Window spectrums of the proposed window for different γ values for $N = 101$ and $\alpha = 2$.

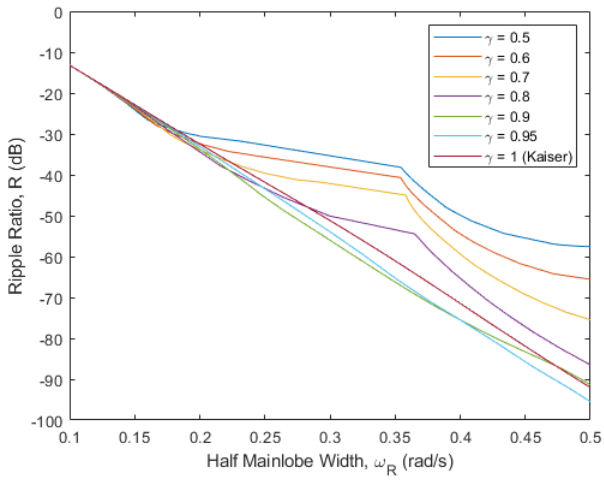

FIGURE 14. Ripple ratio-half mainlobe width characteristics of the proposed window for different γ values (<1) at $N = 51$.

by changing α values at a fixed window length ($N = 51$). Figure 14 and Figure 15 show these relations for $\gamma < 1$ and $\gamma > 1$, respectively. Also, the relation for the Kaiser window is included for the comparison purpose.

It is seen that the proposed window may perform better window spectral characteristics than the Kaiser window only for $\gamma < 1$ as shown in Figure 14, while presenting always worse characteristics for $\gamma > 1$ as shown in Figure 15. Data belonging to Figure 14 are summarized in Table 3 that shows in which main lobe width and ripple ratio ranges the proposed window gives better ripple ratio values than the Kaiser window for a fixed mainlobe width at different γ values. It also shows that the ranges increase as γ value approaches 1. In addition, the maximum difference in ripple ratio occurring for each range is given in the table.

B. FIR FILTER DESIGN USING THE PROPOSED WINDOW

To observe the effects of the independent parameters of the proposed window on the filter spectral characteristics,

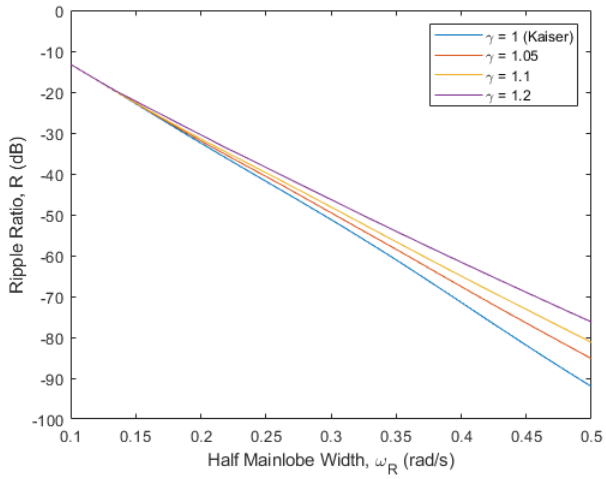

FIGURE 15. Ripple ratio-half mainlobe width characteristics of the proposed window for different γ values (>1) at $N = 51$.

TABLE 3. Window spectral ranges where the proposed window is better than the kaiser window for different γ values.

γ	N	Range for ω_R (rad/s)	Range for R (dB)	Maximum Difference in R (dB)
0.50	51	$0.1001 < \omega_R < 0.1828$	$-13.25 < R < -29.20$	1.67
0.60	51	$0.1001 < \omega_R < 0.1983$	$-13.25 < R < -32.15$	1.86
0.70	51	$0.1001 < \omega_R < 0.2238$	$-13.25 < R < -36.96$	2.06
0.80	51	$0.1001 < \omega_R < 0.2821$	$-13.25 < R < -47.82$	2.35
0.90	51	$0.1001 < \omega_R < 0.4734$	$-13.25 < R < -86.73$	5.20
0.95	51	$0.1001 < \omega_R < 0.5000$	$-13.25 < R < -95.50$	3.81

TABLE 4. Filter spectral values for digital filters designed with proposed window at different filter lengths and α values for $\gamma = 0.9$.

	N = 51		N = 71		N = 101	
α	$\Delta\omega$ (rad/s)	A_s (dB)	$\Delta\omega$ (rad/s)	A_s (dB)	$\Delta\omega$ (rad/s)	A_s (dB)
0	0.1234	20.98	0.0883	20.99	0.0619	20.98
2	0.2096	31.68	0.1498	31.64	0.1049	31.62
4	0.4256	57.37	0.3035	57.24	0.2123	57.17
6	0.7078	85.14	0.5052	84.59	0.3535	84.32
8	0.8967	98.51	0.6397	98.34	0.4474	98.26

low-pass digital filters are designed for different values of the parameters N, α, and γ.

First, for a fixed value of γ, the filter spectrums are obtained for different α values at filter lengths of $N = 51$, 71, and 101 as shown in Figure 16, Figure 17, and Figure 18, respectively. The filter spectral parameters from these figures are summarized in Table 4. It shows that increasing α results in a wider transition width and a higher minimum stopband attenuation for each filter length. This is an expected result from Table 1 because the transition width and minimum stopband attenuation are related with mainlobe width and ripple ratio parameters, respectively.

Secondly, for a fixed value of α, the filter spectrums are obtained for different γ values at filter lengths of $N = 51$, 71, and 101 as illustrated in Figure 19, Figure 20, and Figure 21, respectively. The filter spectral parameters from these figures

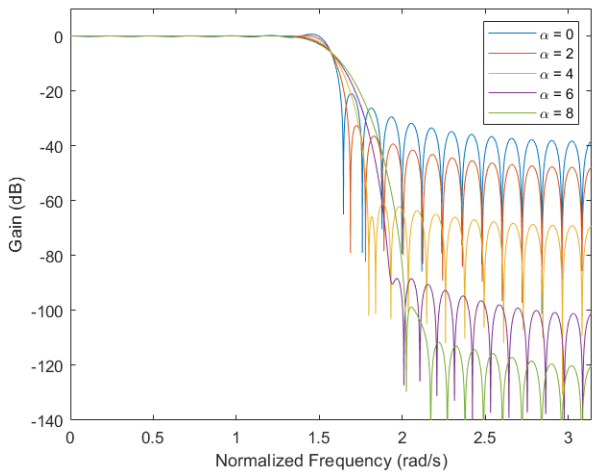

FIGURE 16. Amplitude spectra of filters designed using the proposed window for different α values for $N = 51$ and $\gamma = 0.9$.

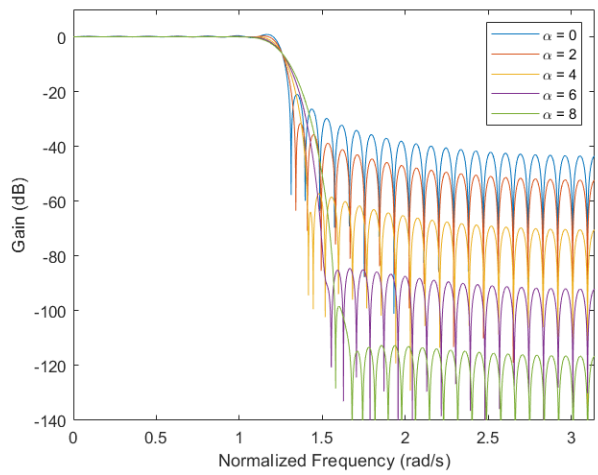

FIGURE 17. Amplitude spectra of filters designed using the proposed window for different α values for $N = 71$ and $\gamma = 0.9$.

TABLE 5. Filter spectral values for digital filters designed with proposed window at different filter lengths and γ values for $\alpha = 2$.

	N = 51		N = 71		N = 101	
γ	$\Delta\omega$ (rad/s)	A_s (dB)	$\Delta\omega$ (rad/s)	A_s (dB)	$\Delta\omega$ (rad/s)	A_s (dB)
0.5	0.6519	56.66	0.4665	56.59	0.3268	56.51
0.6	0.4063	45.59	0.2901	45.52	0.2031	45.46
0.7	0.2863	41.80	0.2046	41.73	0.1433	41.68
0.8	0.2380	35.71	0.1701	35.67	0.1191	35.64
0.9	0.2096	31.68	0.1498	31.64	0.1049	31.62

are summarized in Table 5. It is seen that an increase in γ results in a narrower transition width and a smaller minimum stopband attenuation for each filter length.

Thirdly, the filter spectrums are obtained for different α and γ values at $N = 51$ as shown in Figure 22, Figure 23, and Figure 24, respectively. The filter spectral parameters from these figures are summarized in Table 6. It shows that increasing α leads to a wider transition width, but the minimum stopband attenuation value increases up to a certain α value and then decreases.

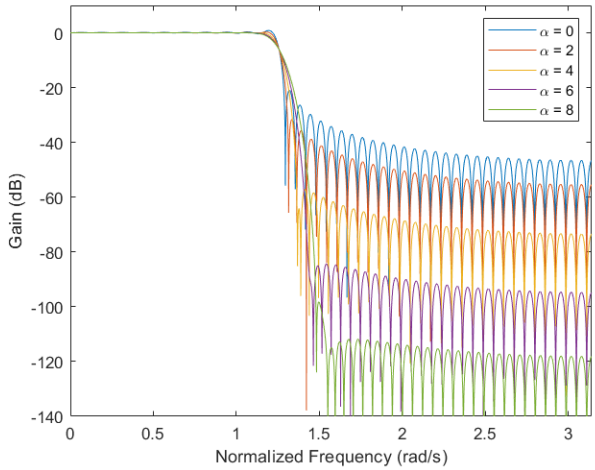

FIGURE 18. Amplitude spectra of filters designed using the proposed window for different α values for $N = 101$ and $\gamma = 0.9$.

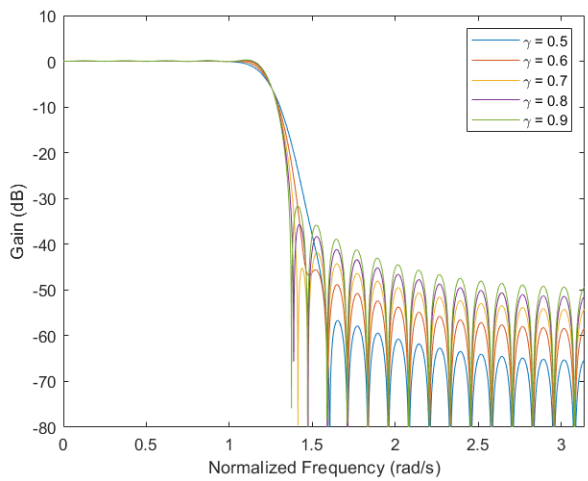

FIGURE 19. Amplitude spectra of filters designed using the proposed window for different γ values for $N = 51$ and $\alpha = 2$.

TABLE 6. Filter spectral values for digital filters designed with proposed window at different γ and α values for $N = 51$.

	$\gamma = 0.6$		$\gamma = 0.7$		$\gamma = 0.8$	
α	$\Delta\omega$ (rad/s)	A_s (dB)	$\Delta\omega$ (rad/s)	A_s (dB)	$\Delta\omega$ (rad/s)	A_s (dB)
0	0.1234	20.98	0.1234	20.98	0.1234	20.98
2	0.4063	45.59	0.2863	41.80	0.2380	35.71
4	1.0174	88.17	0.8374	85.79	0.6552	69.65
6	1.0669	80.86	1.0251	84.97	0.9304	97.33
8	1.0835	79.94	1.0617	81.10	1.0107	85.91

C. GENERAL FILTER SPECTRAL CHARACTERISTICS OF THE PROPOSED WINDOW AND COMPARISON WITH KAISER WINDOW

In previous section, digital filters are designed by using the proposed window for specific values of adjustable parameters. In this section, to better understand the impact of the new independent parameter, γ, on the filter's spectral characteristics, the relationship between the minimum stopband attenuation and the normalized filter transition width,

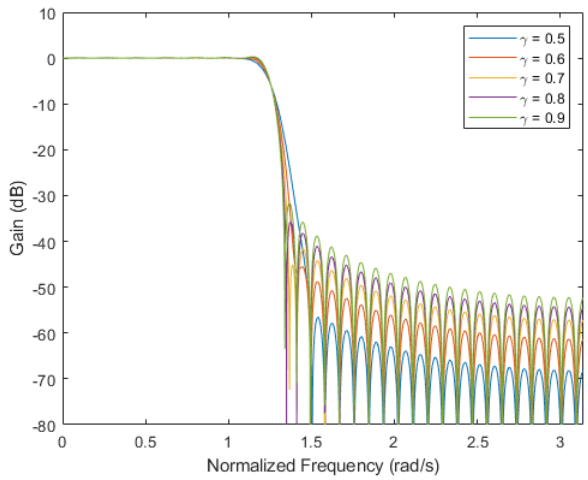

FIGURE 20. Amplitude spectra of filters designed using the proposed window for different γ values for $N = 71$ and $\alpha = 2$.

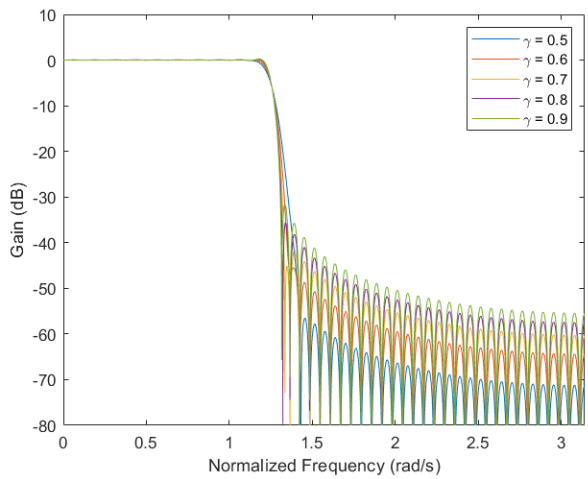

FIGURE 21. Amplitude spectra of filters designed using the proposed window for different γ values for $N = 101$ and $\alpha = 2$.

as defined in (37), is examined for the proposed window at different γ values. These relationships are obtained by varying the α values while keeping the filter length fixed at $N = 51$.

$$D_f = \frac{\Delta\omega(N-1)}{\omega_s}. \tag{37}$$

Figure 25 and Figure 26 show these relationships for $\gamma < 1$ and $\gamma > 1$, respectively. Also, the relation for Kaiser window is included for comparison purpose. It is seen that the proposed window may perform better filter spectral characteristics than the Kaiser window only for $\gamma < 1$ as shown in Figure 25, while presenting always worse characteristics for $\gamma > 1$ as shown in Figure 26. These results are consistent with those obtained from Figure 14 and Figure 15. Data belonging to Figure 25 are summarized in Table 7 that shows in which normalized filter transition width and minimum stopband attenuation ranges the proposed window gives better minimum stopband attenuation values than the Kaiser window for a fixed filter transition width at different γ values. It can be seen from the Table 7 that the ranges increase as the γ value approaches 1. These results are consistent with the results obtained from Table 3. In addition, the maximum difference in minimum stopband attenuation occurring for each range is given in the table.

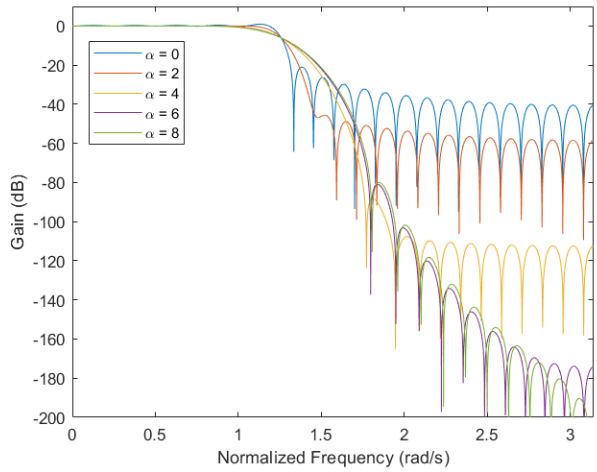

FIGURE 22. Amplitude spectra of filters designed using the proposed window for different α values at $\gamma = 0.6$ and $N = 51$.

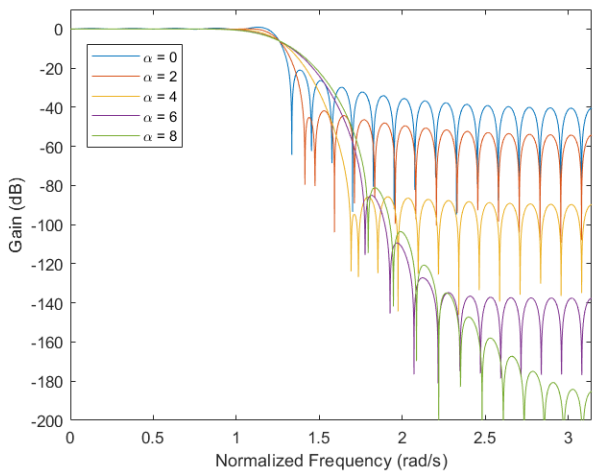

FIGURE 23. Amplitude spectra of filters designed using the proposed window for different for different α values at $\gamma = 0.7$ and $N = 51$.

TABLE 7. Filter spectral ranges where the proposed window is better than the kaiser window for $\omega_{ct} = 0.4\pi$ rad/s at different γ values.

γ	N	Range for D_f (rad/s)	Range for A_s (dB)	Maximum Difference in A_s (dB)
0.50	51	$0.1234 < D_f < 2.1583$	$20.98 < A_s < 38.65$	2.27
0.60	51	$0.1234 < D_f < 2.3282$	$20.98 < A_s < 41.10$	2.48
0.70	51	$0.1234 < D_f < 2.5668$	$20.98 < A_s < 44.51$	2.33
0.80	51	$0.1234 < D_f < 2.9578$	$20.98 < A_s < 50.07$	2.33
0.90	51	$0.1234 < D_f < 4.1343$	$20.98 < A_s < 66.76$	2.32
0.95	51	$0.1234 < D_f < 5.8333$	$20.98 < A_s < 91.98$	2.29

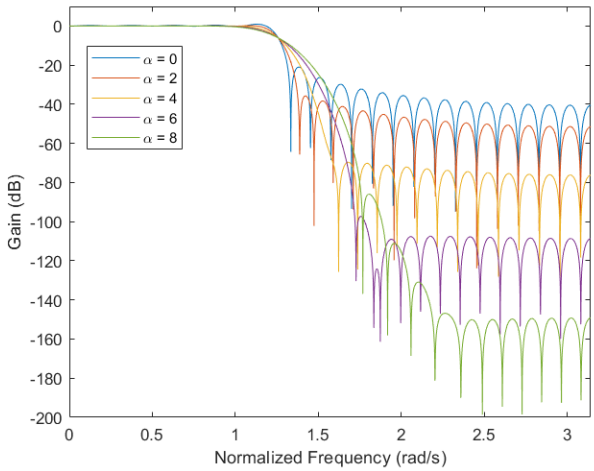

FIGURE 24. Amplitude spectra of filters designed using the proposed window for different α values at $\gamma = 0.8$ and $N = 51$.

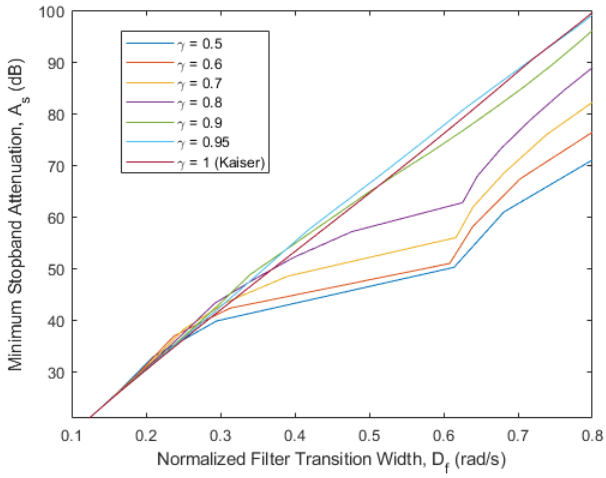

FIGURE 25. Minimum stopband attenuation-filter transition width characteristics of the proposed window for different γ values (<1) and comparison with kaiser window at $N = 51$.

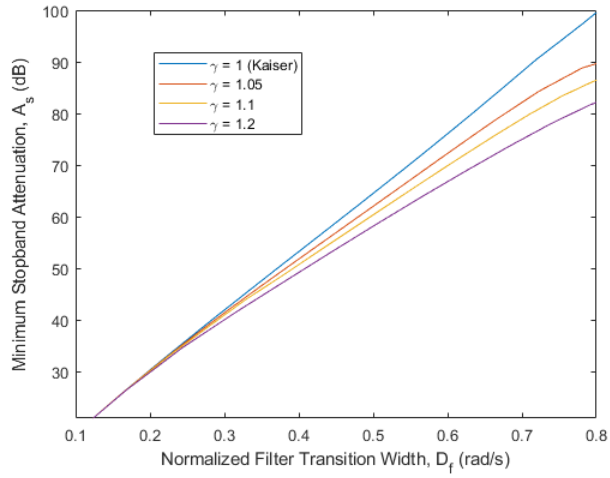

FIGURE 26. Minimum stopband attenuation-filter transition width characteristics of the proposed window for different γ values (>1) and comparison with kaiser window at $N = 51$.

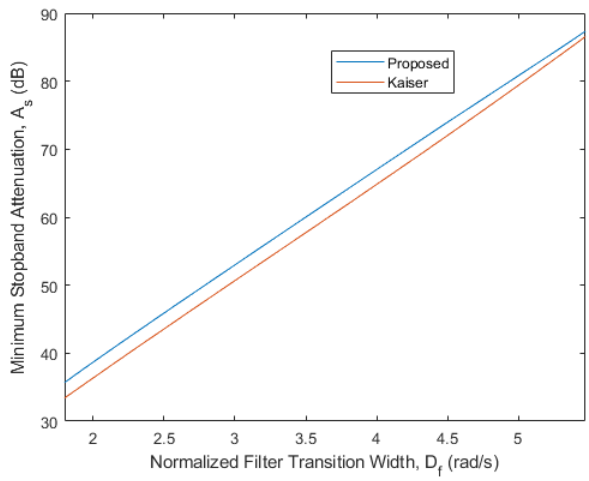

FIGURE 27. Minimum stopband attenuation-filter transition width characteristics comparison between the proposed window and kaiser window at $N = 51$.

Since the proposed window has 3 independent variables, the highest minimum stopband attenuation value can be found for each filter transition width by selecting appropriate α and γ values at a fixed filter length. Figure 27 shows the relation for filter spectral values of the digital filters designed by the proposed window with optimum α and γ values found empirically for N = 51 and compares this relation with that of the digital filters designed by Kaiser window. As can be seen from the figure, the filters designed with the proposed window give better results than the filters designed with the Kaiser window with a numerical range of values from 0.82 dB to 2.48 dB. The relationship between the minimum stopband attenuation values of the filters designed with the proposed window and the corresponding α and γ parameters that produce these values is given in Figure 28.

D. FILTER DESIGN COMPARISON EXAMPLES WITH THE WINDOWS IN LITERATURE

In this section, in addition to Kaiser window, the comparisons are extended to include other well-known adjustable windows such as Saramaki and Dolph-Chebyshev windows. Three design comparison examples with different filter lengths and filter transition widths are provided for a cutoff frequency of $\omega_c = 0.4\pi$ rad/sample.

1) FILTER DESIGN COMPARISON EXAMPLE-I

As a first example, design FIR digital filters to satisfy the given prescribed filter specifications of $\Delta\omega = 0.4000$ rad/sample and $N = 51$.

The amplitude spectrums of the designed filters are given in Figure 29, Figure 30, and Figure 31. The obtained data for $\Delta\omega$ and A_s from these figures are summarized in Table 8.

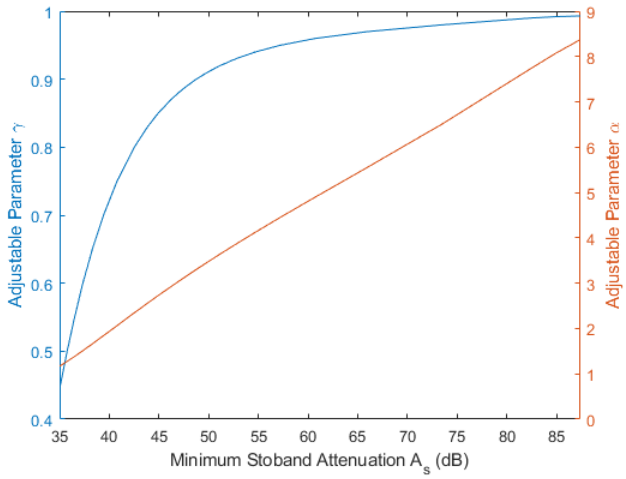

FIGURE 28. Relationship between γ and α parameters with the minimum stopband attenuation of the digital filters using the proposed window at $N = 51$.

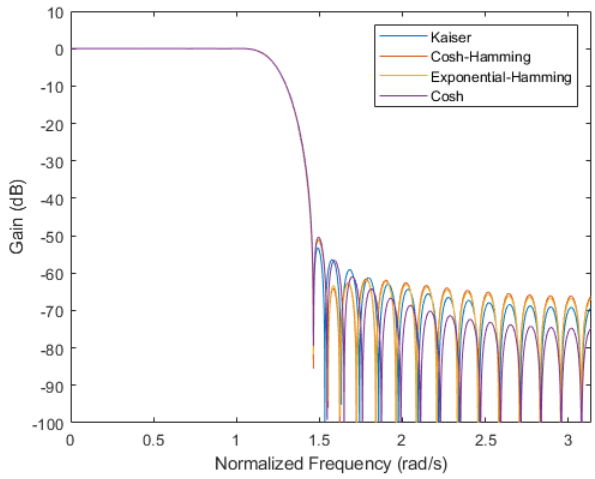

FIGURE 30. Amplitude spectra of the second four filters that give the best results for filter design example I at $N = 51$.

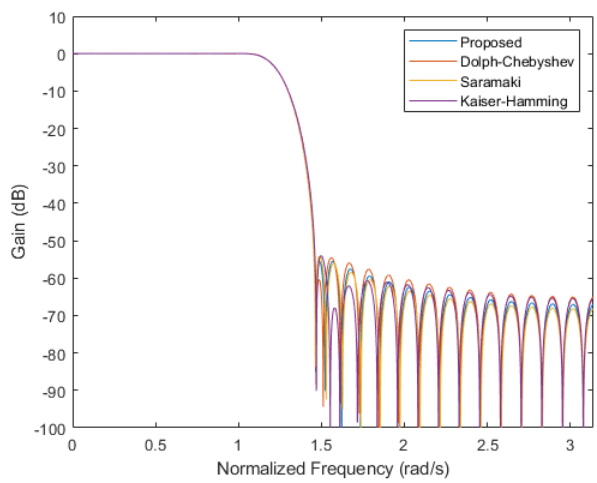

FIGURE 29. Amplitude spectra of the first four filters that give the best results for filter design example I at $N = 51$.

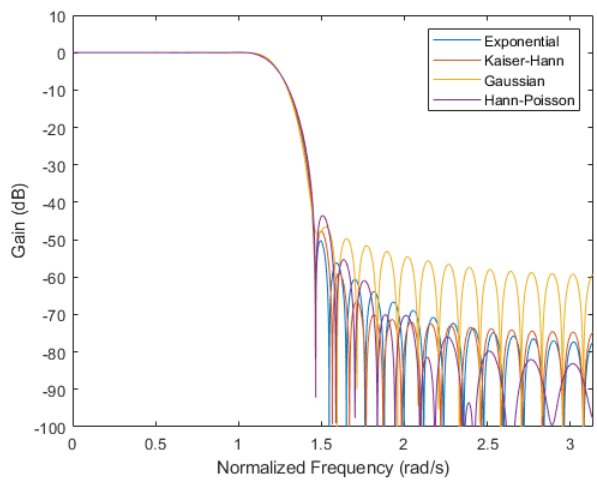

FIGURE 31. Amplitude spectra of the third four filters that give the best results for filter design example I at $N = 51$.

TABLE 8. Results for filter design comparison example I.

Window Function	N	Adjustable Parameters	$\Delta\omega$ (rad/s)	A_s (dB)
Proposed	51	$\gamma = 0.9438$, $\alpha = 4.239$	0.4000	55.57
Dolph-Chebyshev [25]	51	$R = 43.59$ dB	0.4000	54.58
Saramaki [27]	51	$\beta = 1.8972$	0.4000	54.12
Kaiser-Hamming [32]	51	$\alpha = 4.573$	0.4000	53.28
Kaiser [26]	51	$\alpha = 4.921$	0.4000	53.26
Cosh-Hamming [29]	51	$\alpha = 4.178$	0.4000	51.21
Exponential-Hamming [31]	51	$\alpha = 4.195$	0.4000	51.01
Cosh [28]	51	$\alpha = 4.518$	0.4000	50.39
Exponential [30]	51	$\alpha = 4.538$	0.4000	50.39
Kaiser-Von Hann [33]	51	$\alpha = 4.775$	0.4000	47.69
Gaussian [19]	51	$\alpha = 2.0633$	0.4000	46.56
Hann-Poisson [20]	51	$\alpha = -0.0044$	0.4000	43.56

TABLE 9. Results for filter design comparison example II.

Window Function	N	Adjustable Parameters	$\Delta\omega$ (rad/s)	A_s (dB)
Proposed	71	$\gamma = 0.972$, $\alpha = 5.560$	0.3500	66.58
Saramaki [27]	71	$\beta = 2.204$	0.3500	64.91
Kaiser [26]	71	$\alpha = 6.08$	0.3500	63.54
Dolph-Chebyshev [25]	71	$R = 51.865$ dB	0.3500	63.43
Kaiser-Hamming [32]	71	$\alpha = 6.785$	0.3500	62.34
Exponential-Hamming [31]	71	$\alpha = 6.386$	0.3500	60.99
Cosh-Hamming [29]	71	$\alpha = 6.385$	0.3500	60.97
Exponential [30]	71	$\alpha = 5.793$	0.3500	59.85
Cosh [28]	71	$\alpha = 5.786$	0.3500	59.77
Kaiser-Von Hann [33]	71	$\alpha = 7.83$	0.3500	57.18
Gaussian [19]	71	$\alpha = 2.15183$	0.3500	47.37
Hann-Poisson [20]	71	$\alpha = 0.2381$	0.3500	40.02

Also, the adjustable parameters used for the windows to design the required filters are listed in the table. It is seen that the proposed window provides the best minimum stopband attenuation, followed by Dolph-Chebyshev, Saramaki, Kaiser-Hamming and Kaiser windows, respectively. The Hann-Poisson window provides the worst result, followed by Gaussian, and Kaiser-Von Hann windows, respectively.

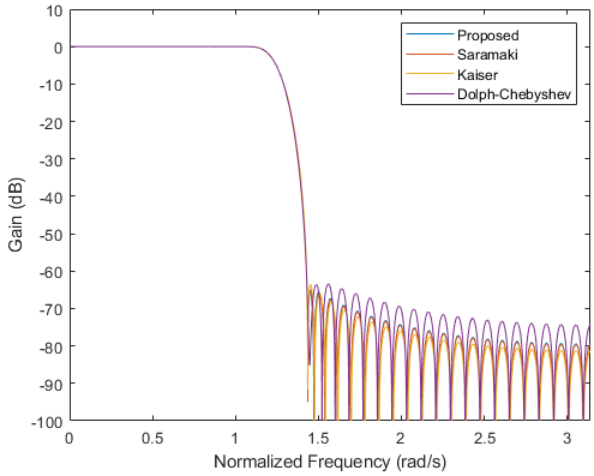

FIGURE 32. Amplitude spectra of the first four filters that give the best results for filter design example II at $N = 71$.

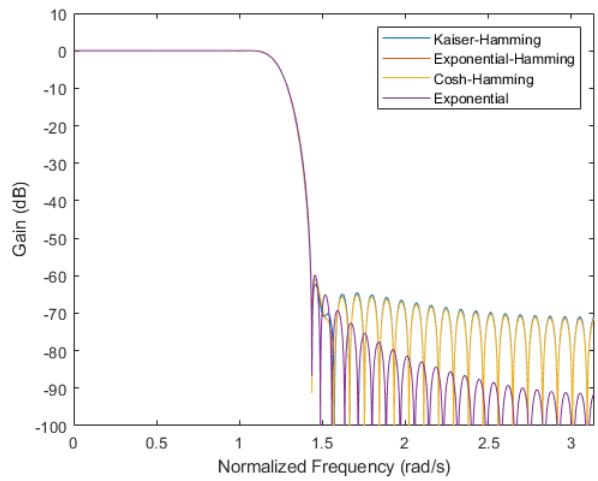

FIGURE 33. Amplitude spectra of the second four filters that give the best results for filter design example II at $N = 71$.

2) FILTER DESIGN COMPARISON EXAMPLE-II

As a second example, design FIR digital filters to satisfy the given prescribed filter specifications of $\Delta\omega = 0.3500$ rad/sample and $N = 71$.

The amplitude spectrums of the designed filters are given in Figure 32, Figure 33, and Figure 34. The obtained data for $\Delta\omega$ and A_s from these figures are summarized in Table 9. From the results, it is seen that the proposed window provides the best minimum stopband attenuation, followed by Saramaki, Kaiser, Dolph-Chebyshev, and Kaiser-Hamming windows, respectively. The Hann-Poisson window provides the worst result, followed by Gaussian, and Kaiser-Von Hann windows, respectively.

3) FILTER DESIGN COMPARISON EXAMPLE-III

As a last example, design FIR digital filters to satisfy the given prescribed filter specifications of $\Delta\omega = 0.3000$ rad/sample and $N = 101$.

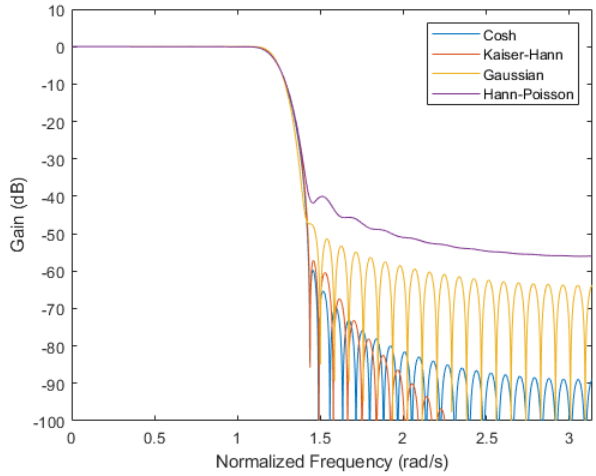

FIGURE 34. Amplitude spectra of the third four filters that give the best results for filter design example II at $N = 71$.

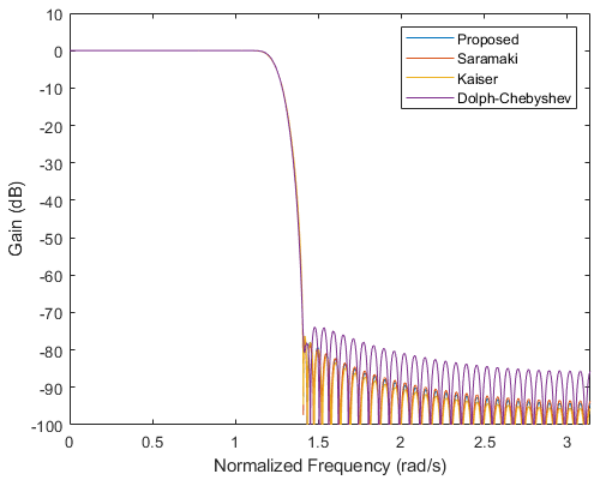

FIGURE 35. Amplitude spectra of the first four filters that give the best results for filter design example III at $N = 101$.

TABLE 10. Results for filter design comparison example III.

Window Function	N	Adjustable Parameters	$\Delta\omega$ (rad/s)	A_s (dB)
Proposed	101	$\gamma = 0.9869$, $\alpha = 7.13$	0.3000	77.91
Saramaki [27]	101	$\beta = 2.597$	0.3000	77.73
Kaiser [26]	101	$\alpha = 7.477$	0.3000	76.35
Dolph-Chebyshev [25]	101	$R = 62.13$ dB	0.3000	73.94
Exponential [30]	101	$\alpha = 7.255$	0.3000	71.89
Cosh [28]	101	$\alpha = 7.258$	0.3000	71.80
Kaiser-Hamming [32]	101	$\alpha = 8.803$	0.3000	64.46
Exponential-Hamming [31]	101	$\alpha = 8.39$	0.3000	64.44
Cosh-Hamming [29]	101	$\alpha = 8.39$	0.3000	64.44
Kaiser-Von Hann [33]	101	$\alpha = 9.838$	0.3000	59.85
Gaussian [19]	101	$\alpha = 2.154$	0.3088	51.36
Hann-Poisson [20]	101	$\alpha = 0.3758$	0.3000	36.33

The amplitude spectrums of the designed filters are given in Figure 35, Figure 36, and Figure 37. The obtained data for $\Delta\omega$ and A_s from these figures are summarized in Table 10. Again, it is seen that the proposed window

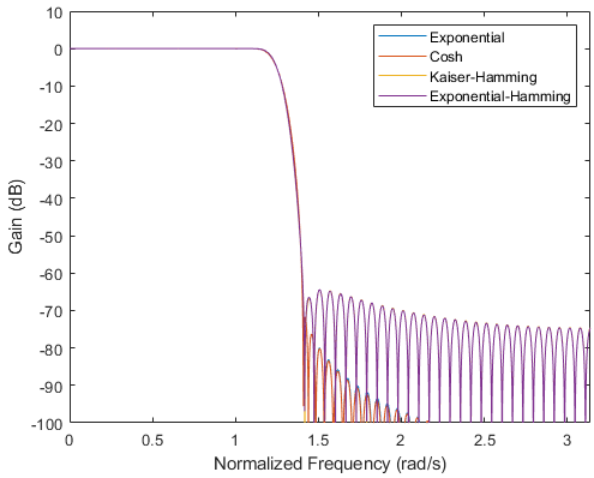

FIGURE 36. Amplitude spectra of the second four filters that give the best results for filter design example III at $N = 101$.

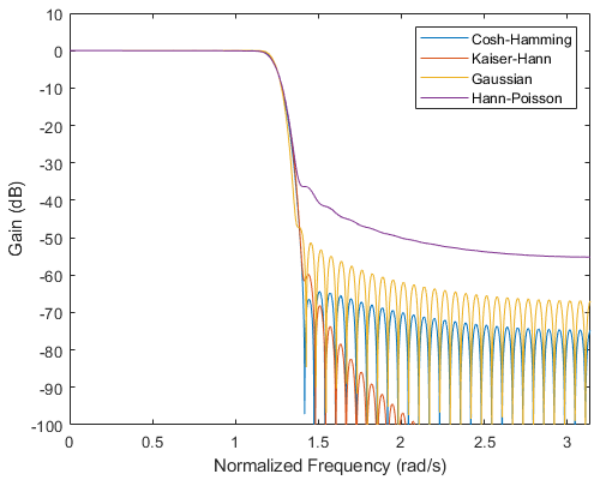

FIGURE 37. Amplitude spectra of the third four filters that give the best results for filter design example III at $N = 101$.

provides the best minimum stopband attenuation, followed by Saramaki, Kaiser, Dolph-Chebyshev, and Exponential windows, respectively. The Hann-Poisson window provides the worst result, followed by Gaussian, and Kaiser-Von Hann windows, respectively. Note that Gaussian window cannot be designed to satisfy the required transition width. Eventhough it is designed with a wider transition width (0.3088 rad/sample), it provides the second worst filtering characteristics.

VI. CONCLUSION

In this study, a new adjustable window function with three independent parameters, namely fractional Kaiser window, is proposed and its performance in digital filter design is examined. The proposed window is based on the fractional modified Bessel function of the first kind with zeroth order. First, the window spectral analysis of the proposed window is performed and then compared with the Kaiser window. It's found that for $\gamma < 1$ the proposed window may outperform the Kaiser window in terms of ripple ratio and mainlobe width characteristics. Then, filter spectral analysis of the proposed window is performed for different filter lengths to observe the effects of the independent adjustable parameters. Subsequently, a general filter design comparison is conducted with the Kaiser window, and the results demonstrate that filters designed using the proposed window achieve better performance, with improvements ranging from 0.82 dB to 2.48 dB. Finally, three specific comparison examples with adjustable windows from the literature are presented to demonstrate that the proposed window offers superior filtering characteristics. Since the proposed window outperforms the Kaiser window and other two-parameter windows, it can be effectively utilized in applications where these windows are commonly applied, such as biomedical processing [52], [53], communications [54], [55], [56], and image processing [57], [58].

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